

NEEDLES, CA, USA

WMO: 723805

Lat: 34.768N

Lon: 114.619W

Elev: 890

StdP: 14.23

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 23179

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	37.4	40.3	-1.8	5.2	65.3	2.3	6.4	66.4	30.2	58.1	27.0	58.4	5.4	240	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	114.7	69.6	112.3	69.3	110.2	69.0	76.9	99.3	75.3	98.4	74.3	98.5	9.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	118.0	86.6	68.4	107.8	88.8	65.8	98.3	91.5	40.7	98.9	39.4	97.9	38.3	98.1	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.5	19.2	DB	31.9	119.0	3.2	2.1	29.6	120.5	27.7	121.8	25.9	122.9	23.6	124.5	(4)
(5)			WB	25.0	78.1	2.8	2.1	23.0	79.7	21.4	80.9	19.8	82.1	17.8	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.8	55.7	59.0	66.6	73.5	82.4	93.2	98.3	96.4	89.4	76.2	64.0	54.0
	DBStd	16.60	5.79	6.32	6.96	6.89	7.25	5.60	4.49	4.47	5.79	6.51	6.80	5.67
	HDD50	52	19	5	0	0	0	0	0	0	0	0	2	26
	HDD65	998	291	184	67	12	0	0	0	0	0	5	98	341
	CDD50	9471	195	256	514	704	1006	1296	1496	1438	1183	812	421	150
	CDD65	4943	2	15	116	266	541	846	1031	973	733	352	67	1
	CDH74	79763	21	188	1354	3344	8160	14925	18695	17090	11460	3942	575	9
	CDH80	54574	1	27	498	1612	5005	10793	14247	12687	7645	1912	147	0
Wind	WSAvg	8.1	8.1	7.9	8.3	9.2	9.0	8.7	8.1	7.6	7.2	7.8	7.6	7.6
Precipitation	PrecAvg	4.6	0.6	0.6	0.5	0.2	0.1	0.0	0.3	0.6	0.5	0.3	0.4	0.4
	PrecMax	9.6	3.6	3.6	3.2	2.8	0.7	0.5	2.2	3.4	4.7	1.7	1.7	2.6
	PrecMin	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	0.9	0.8	0.7	0.6	0.2	0.1	0.5	0.9	0.9	0.4	0.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	83.3	94.2	101.0	109.3	118.2	118.9	115.4	110.8	100.4	88.5	75.3
		MCWB	52.5	55.0	59.8	61.0	66.9	67.0	71.1	70.3	68.8	65.1	59.0	55.1
	2%	DB	72.9	79.6	89.2	96.7	105.8	113.4	115.4	113.1	107.9	96.6	84.2	71.1
		MCWB	51.1	53.8	57.6	59.9	64.0	65.9	70.7	70.8	68.1	63.2	57.0	51.8
	5%	DB	69.8	75.7	85.5	92.8	102.3	110.9	113.0	110.9	105.8	93.4	80.8	67.6
		MCWB	50.0	52.2	56.2	58.4	62.7	65.9	70.6	70.3	68.0	61.9	56.1	50.2
	10%	DB	66.7	71.9	82.0	89.0	99.0	108.3	110.4	108.6	103.3	90.3	76.8	64.4
		MCWB	48.9	51.0	54.7	57.2	61.7	65.5	70.1	70.2	67.5	60.9	54.3	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	59.3	61.8	65.0	70.9	73.2	78.6	79.3	76.0	69.8	62.2	57.5
		MCDB	66.3	72.6	87.6	91.7	104.0	105.9	107.0	101.8	95.8	86.5	79.8	67.6
	2%	WB	55.4	56.6	59.2	61.5	66.5	70.8	76.5	77.4	73.9	67.2	59.4	55.3
		MCDB	64.5	69.7	83.3	89.8	99.6	105.6	100.1	98.6	95.1	87.1	79.5	65.8
	5%	WB	53.2	54.5	57.3	59.9	64.2	68.8	75.2	75.8	72.5	65.2	57.3	53.0
		MCDB	64.6	69.4	81.9	87.6	97.9	104.6	98.9	98.3	94.8	87.7	75.9	63.7
	10%	WB	51.1	52.8	55.8	58.4	62.4	67.2	74.2	74.6	71.0	63.2	55.6	50.7
		MCDB	63.0	68.7	78.8	85.7	96.0	103.7	99.4	98.0	94.7	85.4	74.2	61.5
Mean Daily Temperature Range	5% DB	MDBR	19.7	21.6	24.5	24.7	25.2	26.0	22.5	22.8	24.3	23.7	21.7	18.6
		MCDBR	23.0	27.1	29.6	29.4	29.5	28.7	25.3	25.1	26.6	26.9	25.4	21.8
		MCWBR	11.4	12.7	12.6	12.4	11.7	10.1	8.9	8.8	10.3	12.0	11.9	10.8
	5% WB	MCDBR	19.3	22.5	26.9	26.4	27.6	25.8	20.8	21.3	22.0	22.7	22.6	18.4
		MCWBR	10.1	10.9	11.7	11.5	12.7	11.5	7.5	7.5	9.2	10.8	11.0	9.7
Clear-Sky Solar Irradiance	taub		0.273	0.276	0.290	0.324	0.335	0.334	0.384	0.379	0.338	0.301	0.277	0.272
	taud		2.598	2.612	2.546	2.422	2.419	2.424	2.394	2.401	2.484	2.564	2.606	2.615
	Ebn at Noon		302	312	314	305	301	299	284	284	292	298	298	296
	Edh at Noon		24	26	30	36	37	37	38	37	32	27	23	22
All-Sky Solar Radiation	RadAvg		1039	1353	1821	2259	2552	2709	2407	2235	1978	1531	1153	926
	RadStd		68	101	80	80	82	89	106	94	71	73	48	63

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	-4.75	N/A	-1.80	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.70	N/A	-3.06	N/A	-0.74	N/A	N/A	N/A	+257	+144	

Nomenclature: See separate page